Soham Biren Katlariwala

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IIT Madras B.S. Degree

Summary

Final-year computer engineering (AI & ML) and health science student working on clinical machine learning: interpretable risk prediction, cross-site generalizability and fairness of medical AI, neuroimaging and EEG-based modelling, and evidence synthesis (systematic reviews and meta-analyses). Author of 7 full papers (4 IEEE and 1 Springer LNCS conference papers, 1 workshop paper, 1 proceedings paper), 15 further accepted conference papers (first-author papers at FIRE 2026, 17.3% acceptance, ICMIB 2026, oral, 20% acceptance, IEEE IECS 2026 and ICMLANT 2026; 3 co-authored IEEE papers at WIECON-ECE 2026, RAAICON 2026 and CSNT 2027; 2 at IEEE BIBE 2026 and 2 at ICTDsC 2026 with Springer LNNS proceedings, 1 first-author paper at each; 1 at ICRPSET 2026, IEEE track; 3 at MetaSphere 2026, as corresponding author), a first-author oral presentation at IC-AICSE 2026 (full paper under review), 1 journal article, 1 journal letter (in press), and 21 accepted conference presentations at TCT, AHA Scientific Sessions, SVIN, CNS, ASH, ACG, AADMD, NUMSTAT and JICIRSIGC, 8 of them as first author (2 published in the *JACC* supplement, 5 in *Circulation*, 3 in the *American Journal of Gastroenterology* supplement, and 2 to be published in *Blood*).

Research Focus and Interests

Focus: uncertainty-aware clinical ML: conformal prediction under distribution shift; Gaussian-process and Bayesian uncertainty. **Also:** neuroimaging and EEG • fairness of medical AI • meta-analysis

Education

Indian Institute of Technology Madras — **India**
Bachelor of Science (BS) in Data Science and Applications, 4-year programme — *2023 – 2027 (expected)*
- **GPA: 3.22/4.0 (ECE/AACRAO)**

Gujarat Technological University — **India**
Bachelor of Engineering (Honours) in Computer Engineering, 4-year programme — *2023 – 2027 (expected)*
- Honours in Artificial Intelligence and Machine Learning **GPA: 3.40/4.0 (ECE/AACRAO)**

University of the People — **USA**
Bachelor of Health Science (Honours), 4-year programme — *2025 – 2027 (expected)*
- **CGPA: 3.88/4.0**

Research Experience

Research Intern — **February 2026 – Present**
MIT Critical Data, Laboratory for Computational Physiology, MIT — *Remote*
- Co-authored SAKSHI, a community-operated platform for accountability in clinical AI deployment, with a 20-member international team (COLM 2026 workshop paper [P5]).
- Co-authored a study of 36 U.S. health AI curricula (arXiv preprint [R2]); 2nd author of its open Zenodo dataset [R3].

Research Intern — **July 2025 – September 2025**
University of California, Berkeley — *United States*
- Co-authored a review on quantum-enhanced fiber-optic sensing using quantum metrology.
- Investigated AI and quantum-enhanced fiber-optic biosensors for real-time, non-invasive diagnostics.
- Analyzed the integration of ML with quantum sensing, mapping challenges and sensing mechanisms.

Research Intern — **June 2025 – July 2025**
The LNM Institute of Information Technology — *India*
- Reviewed foundational and recent literature on quantum entanglement, focusing on SVD in bipartite systems.
- Studied entanglement entropy and quantum state separability to support theoretical development.
- Implemented data preprocessing and visualization workflows in Python for ML-based quantum analysis.

Research Intern — **January 2025 – August 2025**
Texas A&M University–Kingsville — *United States*
- Developed an eigenvector-centrality-based pruning method to improve fine-tuning efficiency in CNNs (ISVC 2025, Springer LNCS [P7]).
- Constructed neuron-similarity graphs to rank neuron importance and pruned low-centrality neurons.
- Evaluated on VGG16, ResNet50 and EfficientNet across Oxford-Flowers102, Caltech101 and TF-Flowers, reporting accuracy and parameter reduction.

Research Intern **June 2024 – September 2024**

University of California, Berkeley (REYES 2024) *United States*

- Developed a BERT-based model for AI-generated text detection, achieving an **F1-score of 0.9436** and outperforming logistic regression and SVM baselines.
- Built NLP preprocessing pipelines (tokenization, normalization, embeddings); sole-authored paper in the *2024 REYES Proceedings* [P6].

Collaborative Clinical AI & Evidence-Synthesis Research **2025 – Present**

Multi-institution collaborations (cardiology, neurology, neurosurgery, gastroenterology) *Remote*

- Led interpretable and fairness-aware ML analyses for coronary artery disease and stroke risk prediction [A3, A4, A12].
- Benchmarked cross-site generalizability of cardiac AI across 5 international catheterization laboratories (1,190 patients), including a domain-adaptation benchmark [A13].
- Led two systematic reviews and meta-analyses on graph neural networks for pre-operative fMRI in glioma and on EEG machine learning for Parkinson's disease and DBS candidate selection [A14, A15].
- Contributed to diagnostic-accuracy meta-analyses (PRISMA, QUADAS-2, PROBAST, bivariate random-effects models) in gastroenterology, dentistry and neurointervention.

Publications

Journal Articles and Letters

[J1] Chandran MM, **Katlariwala SB**, Morker JK. Beyond target trial emulation: Residual confounding in a statin–colorectal cancer study [Letter]. *The American Journal of Medicine*. 2026. **Accepted, article in press.** doi:10.1016/j.amjmed.2026.09.039 (DOI registration pending)

[J2] Khan R, Khan A, Khan T, Mubashir H, **Katlariwala SB**, Ali W. Rising mortality in elderly men with prostate cancer and hypertensive comorbidity: A two-decade epidemiological analysis with projections to 2035. *Journal of Saidu Medical College*. 2026;16(3s). doi:10.52206/jsmc.2026.16.3s.1716

Peer-Reviewed Conference, Workshop and Proceedings Papers

[P1] **Katlariwala SB**, Gandhi VC, Patel N, Parmar D, Desai A. TriBoost and beyond: Advanced machine learning approaches for diabetes risk prediction. *2025 International Conference on Electronics and Renewable Systems (ICEARS)*, IEEE, 2025, pp. 1780–1785. doi:10.1109/ICEARS64219.2025.10941310

[P2] Arora P, **Katlariwala SB**, Raj M, Moradi M, Sahu S. Evaluation of deep learning architectures for network intrusion detection. *2026 1st International Conference on Advancing Sustainable Solutions through Technologies (ICASST)*, IEEE, 2026, pp. 1–6. doi:10.1109/ICASST68917.2026.11484132

[P3] Gandhi VC, Panchal K, Lakhtaria J, Gami CC, Gameti N, **Katlariwala SB**. Hybrid deep learning model: Weather classification using ResNet-50 and SVM. *2025 International Conference on Sustainable Technologies for Humanity and Smart World (HSWTech)*, IEEE, 2025, pp. 1–7. doi:10.1109/HSWTech64936.2025.11277422

[P4] Srividhya R, Gandhi VC, D A, Rajesh S, **Katlariwala SB**, Panchal K. YOLOv5 with SVM-aided post-classification for high-precision skin lesion detection. *2025 International Conference on Sustainable Technologies for Humanity and Smart World (HSWTech)*, IEEE, 2025, pp. 1–6. doi:10.1109/HSWTech64936.2025.11278157

[P5] Al-Louzi R, Ali HN, Bank AMC, Celi LA, Cocucci C, Gameiro R, Gorijavolu R, Jaiswal N, Koumbia M, Kasimbazi HC, **Katlariwala SB**, Nazarian N, Lucas MM, Madapati K, Mikolutskaya S, Mwavu R, Patel MS, Rojas Sillery RI, Nanyonjo J, Xiang AQ. SAKSHI: A community-operated investigative platform for accountability in clinical AI deployment. *COLM 2026 Workshop on LLM/VLM Deployment Opportunities and Risks in Healthcare*, 2026. openreview.net/forum?id=gnMnqG08xo

[P6] **Katlariwala SB**. BERT-based detection of AI-generated text for content verification. *2024 REYES Proceedings*, Old Dominion University Digital Commons, 14. digitalcommons.odu.edu/reyes-2024/14

[P7] Chowdhury S, **Katlariwala SB**, Kashyap D. Effective fine-tuning with eigenvector centrality based pruning. In: *Advances in Visual Computing (ISVC 2025)*, Lecture Notes in Computer Science, Springer, 2026, pp. 3–15. doi:10.1007/978-3-032-14492-8_1

Accepted and Presented Conference Papers (Proceedings Forthcoming or Under Review)

[C1] Aryan R, Banerjee S, Roy N, Jana S, **Katlariwala SB**. Domain-adversarial feature alignment for unsupervised cross-domain segmentation transfer. *IEEE CSNT 2027*, Khajuraho, India. **Accepted (oral).**

[C2] **Katlariwala SB**, Batabyal D, Roy N, Mallipeddi R. Green's-function Gaussian processes for structural reliability. *ICMIB 2026*. **Oral (20% acceptance); Springer (forthcoming).**

[C3] **Katlariwala SB**, Roy N, Moradi M. Script gap in the judge: Script-induced bias in LLM-generated relevance judgments for Indic retrieval evaluation. *Forum for Information Retrieval Evaluation (FIRE 2026)*. **Regular paper, accepted (17.3% acceptance).**

[C4] **Katlariwala SB**, Kundu S, Sarkar S, Mukherjee S, Mallipeddi R. Interpretable terms, unstable forecasts. *ICMLANT 2026*. **Accepted.**

[C5] Lincoln SS, **Katlariwala SB**, Jothilingam K. Confusion-aware conformal abstention for targeted deferral in clinical differential diagnosis. *ICRPSET 2026*, Rajshahi, Bangladesh. **Accepted, IEEE track.**

[C6] **Katlariwala SB**, Roy N, Kundu S, Adedigba A, Mallipeddi R. Does shift-aware conformal correction help? An audit-gated study on cross-site breast ultrasound. *IEEE International Conference on Bioinformatics and Bioengineering (BIBE 2026)*. **Full paper accepted.**

[C7] Roy N, **Katlariwala SB**, Kundu S, Adedigba A, Mallipeddi R. CLEAR-Net: Adaptive denoising for robust polyp segmentation. *IEEE BIBE 2026*. **Full paper accepted.**

[C8] **Katlariwala SB***, Jothilingam K, Muhammed SS. SHAP-guided iterative feature pruning (SIFP): Stable gene selection for breast cancer subtype classification. *ICTDsC 2026*, Siliguri, India. **Oral; Springer LNNS (forthcoming).**

[C9] Patel JD, **Katlariwala SB***, Chauhan JA, Patel NI, Raval DK, Prajapati CD. Characterising digital twin fidelity under real 5G network impairments via autoregressive Kalman filtering. *ICTDsC 2026*, Siliguri, India. **Oral; Springer LNNS (forthcoming).**

[C10] Muna FH, **Katlariwala SB**, Lincoln SS, Johra I, Tisha NF. Conformal-guaranteed actionable recourse for clinical risk prediction: Toward trustworthy AI for eating-disorder screening and beyond. Paper 488. *IEEE WIECON-ECE 2026*, Dhaka, Bangladesh. **Presented; IEEE Xplore (forthcoming).**

[C11] **Katlariwala SB**, Saeed A, Khati HK, Roy N, Moradi M. Questioning the benchmarks: Hidden circularity in AQ-10 autism screening datasets. *International Conference on AI in Computational Science and Engineering (IC-AICSE 2026)*, Everest Engineering College, Lalitpur, Nepal, Oct 2026. **Oral presentation; full paper under review.**

[C12] Lincoln SS, **Katlariwala SB**, Muna FH. Whose Cuisine? Cultural bias in digital eating support systems. Paper 552. *IEEE RAAICON 2026*. **Accepted.**

[C13] **Katlariwala SB***, Patel JD, Chauhan JA, Patel NI, Raval DK, Prajapati CD. Anatomical GraphSAGE-enhanced CNN ensemble for paediatric bone age estimation with Monte Carlo dropout. Paper 109. *Global Conference on Cloud Computing, Virtualization & Metaverse (MetaSphere 2026)*, IEM, Kolkata, India, Aug 2026.

[C14] **Katlariwala SB***, Chauhan JA, Patel JD, Patel NI, Raval DK, Prajapati CD. Temporal cart memory and billing verification: A three-component framework for vision-based smart trolley systems. Paper 108. *MetaSphere 2026*, IEM, Kolkata, India, Aug 2026.

[C15] Chauhan JA, Patel JD, Patel NI, **Katlariwala SB***. Criterion-level automated scoring of the FMS deep squat using interpretable 2D pose geometry. Paper 203. *MetaSphere 2026*, IEM, Kolkata, India, Aug 2026.

[C16] **Katlariwala SB**, Solanki DN, Mehta D, Gandhi VC, Moradi M, Chauhan A. Physics-aware interpretable ensembles for concrete strength prediction with uncertainty. *IEEE IECS 2026*. **Presented; IEEE Xplore (forthcoming).**

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Preprints and Datasets

[R1] **Katlariwala SB**, Saeed A, Khati HK, Roy N, Moradi M. Questioning the benchmarks: Hidden circularity in AQ-10 autism screening datasets. *medRxiv*. 2026. doi:10.64898/2026.11.01.26367003

[R2] Giancotti R, Santucci F, Anjum A, de Villa M, Mitra R, **Katlariwala SB**, Mishra V, Celi LA. Data provenance and ethics in the curricula of U.S. health AI education programs. *arXiv preprint* arXiv:2610.39606 [cs.CY], 2026. doi:10.48550/arXiv.2610.39606

[R3] Giancotti R, **Katlariwala SB**, Anjum A, Santucci F, Mitra R, de Villa M, Mishra V, Celi LA. Data provenance and ethics in the curricula of U.S. health AI education programs [Dataset]. *Zenodo*. 2026. doi:10.5281/zenodo.22936034

Conference Presentations (21)

[A1] Srinivasan V, Bhavanam S, Zafarullah M, Sajid H, Chandran MM, **Katlariwala SB**, et al. Diagnostic accuracy of artificial intelligence in differentiating iron deficiency anemia from thalassemia trait: A systematic review and meta-analysis. Abstract abs26-12948. *American Society of Hematology (ASH) 2026 Annual Meeting*; to appear in *Blood*.

[A2] Sakalabaktula KSK, Hayat M, Rehman L, Kuraku DR, Latif B, Adel A, **Katlariwala SB**, et al. Diagnostic accuracy of artificial intelligence/machine learning assisted flow cytometry for measurable residual disease detection in leukemia: A systematic review and meta-analysis. Abstract abs26-13932. *American Society of Hematology (ASH) 2026 Annual Meeting*; to appear in *Blood*.

[A3] **Katlariwala SB**, Jothilingam K, Gandhi V, Moradi M. Trajectory-informed stroke risk prediction using disease duration and glycemic burden in a nationally representative US adult sample. Abstract 26-A-1291. *Society of Vascular and Interventional Neurology (SVIN) 2026 Annual Meeting*, Las Vegas, NV, Nov 2026. **Moderated poster.**

[A4] **Katlariwala SB**, Jothilingam K, Gandhi V, Moradi M. Logistic regression outperforms deep ensembles for tabular stroke risk prediction: A fairness-aware benchmarking study. Abstract 26-A-1299. *SVIN 2026*. Poster.

[A5] Jothilingam K, **Katlariwala SB**, Gandhi V, Moradi M. Patient-agnostic wearable activity recognition via cross-subject contrastive learning: Toward post-stroke monitoring. Abstract 26-A-1415. *SVIN 2026*. E-poster.

[A6] Jothilingam K, Zafarullah M, **Katlariwala SB**, Kumar A, Ahmed F, Tabassum H, Khan M. Transradial versus transfemoral access in posterior circulation endovascular interventions: A systematic review and meta-analysis. Abstract 26-A-1303. *SVIN 2026*. E-poster.

[A7] **Katlariwala SB**, Alkashef A, Chandran MM, Pineda Guerrero A, Gandhi V, Sivasubramanian D. Bedside risk stratification in the cardio-metabolic intensive care unit: Machine learning applications for comorbid heart failure and diabetes. Abstract 16575. *AHA Scientific Sessions 2026*; *Circulation* 2026;154(Suppl 1). doi:10.1161/circ.154.suppl_1.16575

[A8] Azhar Z, Abbas N, Khadim MD, Hamid M, **Katlariwala SB**, et al. Baseline LDL cholesterol and clinical outcomes in older adults: Reassessing the cholesterol paradox. A systematic review and meta-analysis. Abstract 18422. *AHA Scientific Sessions 2026*; *Circulation* 2026;154(Suppl 1). doi:10.1161/circ.154.suppl_1.18422

[A9] Ghauri HF, Amjad S, **Katlariwala SB**, et al. Machine learning versus conventional risk models for predicting futility and adverse clinical outcomes following transcatheter aortic valve replacement: A systematic review and meta-analysis. Abstract 17460. *AHA Scientific Sessions 2026*; *Circulation* 2026;154(Suppl 1). doi:10.1161/circ.154.suppl_1.17460

[A10] Hayat M, Tahir B, Arif A, **Katlariwala SB**, et al. Diagnostic accuracy of artificial intelligence-enabled electrocardiography for detecting occlusion myocardial infarction: A systematic review and meta-analysis. Abstract 15546. *AHA Scientific Sessions 2026*; *Circulation* 2026;154(Suppl 1). doi:10.1161/circ.154.suppl_1.15546

[A11] Latif B, Hayat M, Rehman L, **Katlariwala SB**, et al. Remote-only follow-up strategies for patients with cardiac implantable electronic devices: A systematic review and meta-analysis of clinical outcomes. Abstract 14440. *AHA Scientific Sessions 2026*; *Circulation* 2026;154(Suppl 1). doi:10.1161/circ.154.suppl_1.14440

[A12] **Katlariwala SB**, Chunchu VA, Gandhi V, et al. Valvular heart disease severity modifies the discriminative power of clinical coronary artery disease predictors: An interpretable machine learning analysis. Abstract TCT-34. *TCT 2026*, San Diego, CA; *JACC* 2026;88(18_Suppl):B14. doi:10.1016/j.jacc.2026.09.077

[A13] **Katlariwala SB**, Mohammed S, Gandhi V, Qamar I, Moradi M. Institutional referral threshold bias limits generalizability of cardiac AI across 5 international catheterization laboratories: A 1,190-patient cross-site analysis with domain adaptation benchmark. Abstract TCT-958. *TCT 2026*, San Diego, CA; *JACC* 2026;88(18_Suppl):B409. doi:10.1016/j.jacc.2026.09.1168

[A14] **Katlariwala SB**, Pineda Guerrero A, Romero Moreno J, et al. Graph neural network analysis of pre-operative resting-state fMRI predicts extent of resection and post-operative neurocognitive outcome in eloquent-area diffuse glioma: A systematic review and meta-analysis. Poster 4657. *Congress of Neurological Surgeons (CNS) 2026 Annual Meeting*, Washington, DC, Oct–Nov 2026. **Presented.**

[A15] **Katlariwala SB**, Pineda Guerrero A, Romero Moreno J, et al. Machine learning of EEG biomarkers for Parkinson's disease subtyping and deep brain stimulation candidate stratification: A systematic review and meta-analysis. Poster 3576. *CNS 2026*. **Presented.**

[A16] **Katlariwala SB**, Patel JD, Chauhan JA, Patel NI, Raval DK, Prajapati CD. Conformal prediction versus classical parametric reliability models for fatigue life: A fairness-corrected statistical comparison. Abstract 09M. *NUMSTAT 2026*, ABV-IIITM Gwalior, India.

[A17] Qamar I, Hayat M, Chandran MM, **Katlariwala SB**, et al. S1142: Computer-aided detection versus computer-aided diagnosis of early Barrett's neoplasia: A systematic review and meta-analysis. *American College of Gastroenterology (ACG) 2026 Annual Scientific Meeting. Am J Gastroenterol.* 2026;121(10S1):S248. doi:10.14309/01.ajg.0001213820.29928.0b

[A18] Mohammed S, Marupuru S, Sikander DM, Chandran MM, **Katlariwala SB**, et al. S3467: Diagnostic precision and operational efficiency of AI-assisted esophagogastroduodenoscopy for gastric neoplasm detection: A systematic review and meta-analysis of randomized controlled trials. *ACG 2026. Am J Gastroenterol.* 2026;121(10S1):S755. doi:10.14309/01.ajg.0001223120.73712.78

[A19] Singh S, Sikander DM, Vijaykumar R, Chandran MM, **Katlariwala SB**, et al. S2917: Artificial intelligence versus conventional prognostic scores for mortality prediction in acute-on-chronic liver failure: A systematic review. *ACG 2026. Am J Gastroenterol.* 2026;121(10S1):S634. doi:10.14309/01.ajg.0001220920.71799.8e

[A20] **Katlariwala SB**, Muna FH. Predicting household food security among climate-vulnerable coastal women in Bangladesh using explainable artificial intelligence. *JICIRSIGC 2026*, Dhaka, Bangladesh. Poster.

[A21] Bisani JT, **Katlariwala SB**, Moradi M, Saxena MD, Mukhtar SF, Gyawali P. AI-driven severity grading of dental caries: A systematic review, meta-analysis and translational readiness assessment. *AADMD One Voice Conference*, 2026. Poster.

Honors and Recognition

- **Moderated poster presentation** (first author), SVIN 2026, Las Vegas (Abstract 26-A-1291).
- **Oral presentation** (first author), IC-AICSE 2026, Lalitpur, Nepal (IEEE Nepal Section).

Technical Skills

Programming & ML: Python, R, SQL, PyTorch, TensorFlow, XGBoost/LightGBM, GNNs, Gaussian processes
Evaluation: SHAP, conformal prediction, calibration, DeLong, bootstrap CIs, meta-analysis (PRISMA, QUADAS-2)